Genomics and Science Dojo 2.0

3-Minute Presentation

Lasso-Logistic Regression and Artificial Neural Network for Prostate Adenocarcinoma Metastasis Prediction and Biomarker Discovery Based on Gene Expression Profile

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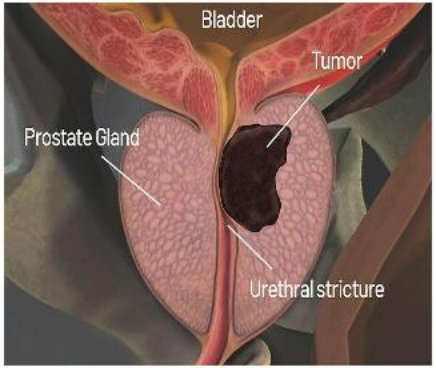
Supported by:

Introduction

Prostate adenocarcinoma / prostate cancer



4th most prevalent cancer among men in Indonesia

Frequently advances to metastasis

Metastasis detection relies on histopathology and CT scans, which are takes time and costly.

Research objective:

1. **To develop a machine learning model** for predicting prostate cancer metastasis using gene expression data.
2. **To identify key biomarkers** linked to prostate cancer metastasis for improved diagnosis and treatment.

Methods

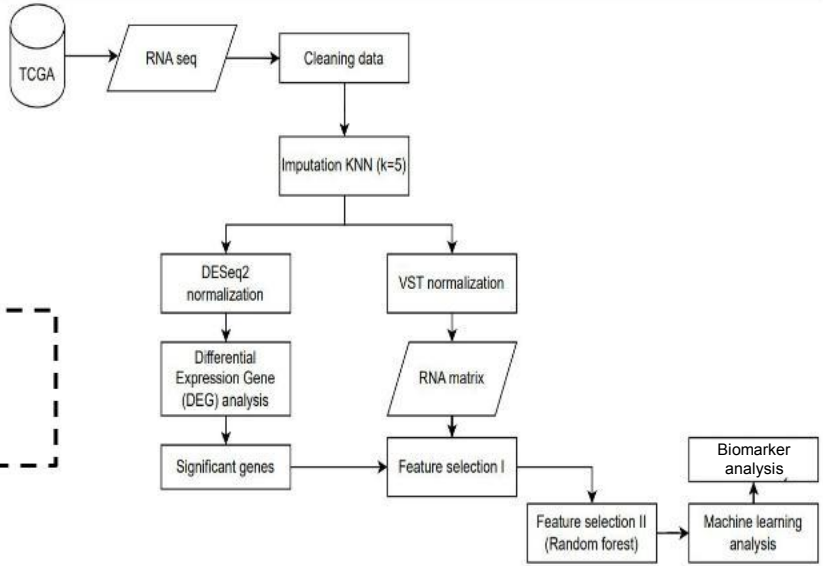


Figure 1.
RNA-seq analysis pipeline, from preprocessing and normalization to feature selection, classification, and biomarker identification.

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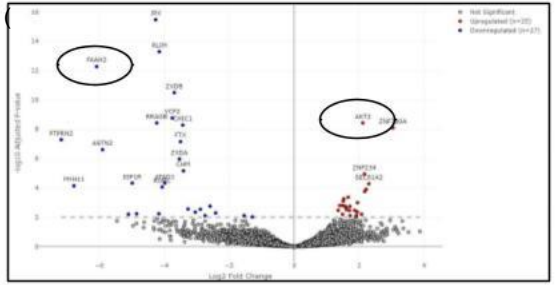
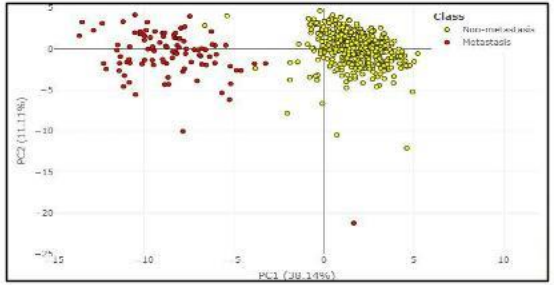
Results

1.Feature selection I : Differential Expression Genes



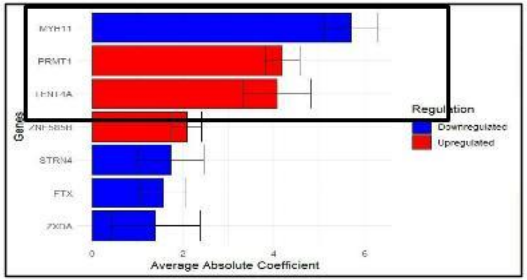
- Several genes are significantly (q<0.01) **down-regulated** (27 genes; such as FAAH) and **up-regulated** (25 genes; such as AKT) in metastatic prostate cancer

2.Data distribution (PCA)



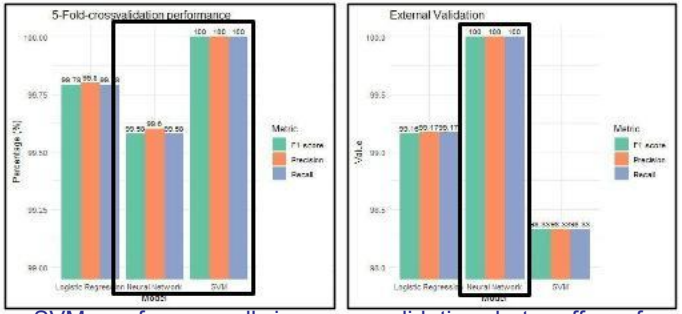
- The PCA plot shows distinct gene expression patterns between **metastatic** (red) and **non-metastatic** (yellow) prostate cancer samples.
- The clear separation of data points indicates **significant differences** in their **gene expression profiles**.

3.Feature selection II : Feature importance



- Trained **Lasso-logistic regression** and **random forest** to assess feature importance, with **logistic regression** showing the best performance (100%)
- Identified seven key genes, selecting three as biomarkers for the next classifier model (**MYH11, PRMT1, TENT4A**)

4.Cross validation & external validation



- SVM performs well in cross-validation but suffers from **overfitting**.
- **External validation** is more important, and the **neural network** is the best at generalizing patterns, making it the **preferred choice**.

5.Biomarker pathway analysis

- **MYH11 (Myosin Heavy Chain 11)** = downregulated in metastasis, leading to **cytoskeleton disorganization** and easier cancer cell detachment from the primary tumor
- **PRMT1 (protein arginine methyltransferase 1)** = upregulated in metastasis, **EMT induction** by regulating E-cadherin and ZEB1, **promotes motility and invasion by β-catenin downregulation**
- **TENT4A** (Terminal Nucleotidyltransferase 4A) = upregulated in metastasis, **regulates mRNA stability and degradation**, potentially influencing cancer progression by modulating oncogene expression and RNA quality control.

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Conclusion

- **Lasso-Logistic Regression** outperformed Random Forest in feature selection, and **Artificial Neural Network** achieved high classification performance during external validation using only three biomarkers.
- The three identified biomarkers **(MYH11, PRMT1, and TENT4A)** are associated with metastasis induction through **cytoskeleton disorganization, EMT promotion, and mRNA stabilization.**

Limitation

- This study relies on RNA-seq data from non-Indonesian patients due to limited available patient data in Indonesia.
- Requires more external validation, which can be achieved by increasing the dataset size.